

A Level Computer Science (9618)

Topical Past Paper Worksheet

Topic 4: Processor Fundamentals

Subtopic 4.1: Central Processing Unit (CPU) Architecture

Concept 4.1.6: Peripheral Ports (USB, HDMI, VGA)

Total Questions: 8

Mark Schemes begin on Page 5

Question 1 | Source: 9618_s25_qp_13 (Q2.c)

- 2 A computer has a processor.
- (c) The computer connects to a monitor using a High Definition Multimedia Interface (HDMI) cable that connects into an HDMI port.

Explain how HDMI provides connection to peripheral devices.

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..... [2]

Question 2 | Source: 9618_w24_qp_11 (Q3.b.ii)

- 3 A computer system has a dual-core Central Processing Unit (CPU).
- (ii) A solid state (flash) memory drive is automatically recognised by the computer when it is plugged into a port in the computer.

Identify an appropriate type of port to connect the solid state memory drive to the computer.

Explain how this port provides an automatic connection.

Port

Explanation

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..... [3]

Question 3 | Source: 9618_w24_qp_12 (Q3.b)

- 3 A student has a computer.

- (b) The student needs to connect the computer to a monitor that has a screen resolution of 2560 × 1600 pixels. The monitor also has built-in speakers.

The computer has a Video Graphics Array (VGA) port and a High Definition Multimedia Interface (HDMI) port.

Explain the benefits of connecting the monitor to the computer using the HDMI port instead of the VGA port.

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..... [4]

Question 4 | Source: 9618_w23_qp_12 (Q8.c.i)

- 8 A computer has an Operating System (OS).

(c) An optical disc reader/writer is connected to the computer.

- (i) Give the name of **one** port that can provide a connection for the optical disc reader/writer.

..... [1]

Question 5 | Source: 9618_w23_qp_13 (Q7.d)

- 7 A laptop computer has Static RAM (SRAM).

A virtual reality headset and a laser printer are connected to the laptop.

- (d) Identify **one** port that could be used to connect the virtual reality headset to the laptop.

Justify your choice.

Port

Justification

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[3]

Question 6 | Source: 9618_s23_qp_12 (Q5.c.i)

5 A student has purchased a new laptop.

(c) Peripherals are connected to the laptop using ports.

(i) A printer is connected to a Universal Serial Bus (USB) port.

Describe how data is transmitted through a USB port.

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..... [1]

Question 7 | Source: 9618_s23_qp_12 (Q5.c.ii)

5 A student has purchased a new laptop.

(c) Peripherals are connected to the laptop using ports.

(ii) A monitor is connected to the laptop using a different type of port.

Identify **one** other type of port that can be used to connect the monitor.

..... [1]

Question 8 | Source: 9618_w22_qp_11 (Q7.e)

(e) State the name of a peripheral device port that provides a physical connection in the computer for each of these peripherals.

3D printer

Monitor [2]

MARK SCHEMES

Mark Scheme for Question 1 | Source: 9618_s25_ms_13 (Q2.c)

2(c)	1 mark each to max 2 e.g. • HDMI transfers both audio and video using a single cable • HDMI has a high bandwidth • Data is transmitted in a stream of uncompressed digital signals • HDMI uses a technology called Transition-Minimized Differential Signalling (TMDS)	2
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Mark Scheme for Question 2 | Source: 9618_w24_ms_11 (Q3.b.ii)

3(b)(ii)	1 mark for correct port. 2 marks for explanation Port: USB / Universal Serial Bus Explanation: • A voltage change occurs when the drive is plugged in • The computer detects this voltage change • The code of the device is transferred to computer • ... the OS finds the code of the device in the list of devices • ... and loads the appropriate device driver	3
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Mark Scheme for Question 3 | Source: 9618_w24_ms_12 (Q3.b)

3(b)	1 mark for each bullet point (max 4) e.g. • HDMI has faster transfer rates than VGA • ... needed due to high resolution / large number of pixels of monitor // HDMI supports the high resolution of the monitor • HDMI supports video and audio transfer between computer and monitor speakers • ... so no separate sound cable is needed unlike VGA • HDMI is digital interface therefore no data is lost in transfer to analogue and back • HDMI is less prone to error/crosstalk/external interference	4
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Mark Scheme for Question 4 | Source: 9618_w23_ms_12 (Q8.c.i)

8(c)(i)	1 mark for each bullet point (max 1) • USB / Universal Serial Bus • HDMI	1
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Mark Scheme for Question 5 | Source: 9618_w23_ms_13 (Q7.d)

7(d)	1 mark for naming a correct port 2 marks for matching justification	3
	• USB • ...has fast data transfer speeds for data (to the headset) • ...is a universal/popular cable // universal standard • HDMI • ...allows video and audio to be transferred (on the same cable) • ...convenience of HDMI as no need for two cables	

Mark Scheme for Question 6 | Source: 9618_s23_ms_12 (Q5.c.i)

5(c)(i)	1 mark for a correct answer	1
	• 1 bit is transferred at a time • Can be synchronous or asynchronous • USB-3 is full duplex and earlier versions are half-duplex	

Mark Scheme for Question 7 | Source: 9618_s23_ms_12 (Q5.c.ii)

5(c)(ii)	1 mark for identification of a suitable port	1
	Examples • HDMI • DisplayPort	

Mark Scheme for Question 8 | Source: 9618_w22_ms_11 (Q7.e)

7(e)	1 mark for each device.	2
	• 3D printer: USB port / COM port • Monitor: HDMI / VGA / USB / DisplayPort	